

Lecture 6: Robust Training (Cont.)

CS 580B1: Trustworthy Machine Learning

Fall 2024

Acknowledgments

Course slides derived from material created by Rajeev Alur and Osbert Bastani at UPenn ([link](#) to their course)

Other relevant courses:

[CS 329T](#) at Stanford

[CS 521](#) at UIUC

[ECE1784H/CSC2559H](#) at U Toronto

Course logistics

Small updates to course schedule

8	Oct 10	Thurs	Introduction to LLMs	(1) Formal Algorithms for Transformers ; (2) (Optional) Attention Is All You Need ; (3) (Optional) The Illustrated Transformer ; (4) (Optional) Simons Institute: How Do Transformers Work?	Due	Proposal Due
9	Oct 15	Tues	The Lord of the Models: The Fellowship of Trust in AI -- Guest Lecture by Dr. Thomas Zimmermann in CSB 130			
9	Oct 17	Thurs	Mechanistic Interpretability	(1) Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 small ; (2) (Optional) Progress Measures for Grokking via Mechanistic Interpretability	Due	

Quick recap

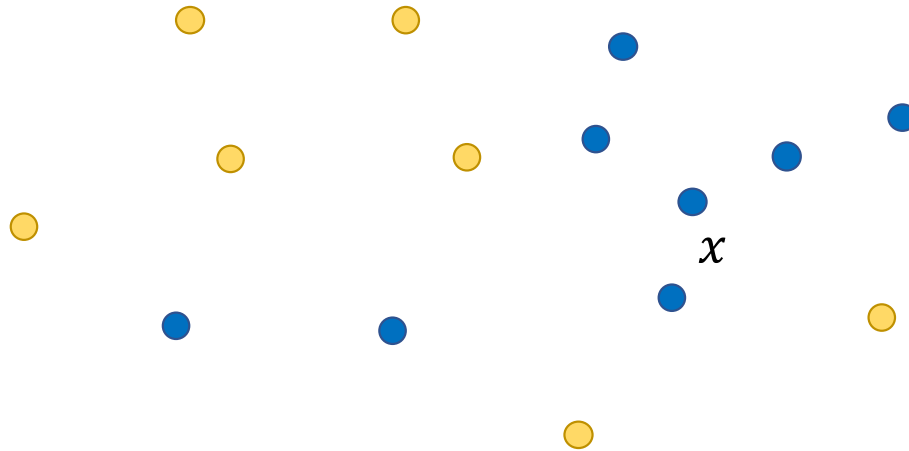
Tuesday:

- Defenses against adversarial examples at test time, inference time, and train time
 - Attack algorithms vs certification algorithms
- Adversarial training:
 - Solving minmax optimization problem
 - Use adversarial attacks during the training loop

Today:

- Certified Adversarial Robustness via Randomized Smoothing by Cohen et al.

Randomized smoothing, informally



- Randomly sample multiple perturbations x' of x
- Compute label $f(x')$ for each perturbed input (where f is a classifier)
- Set $g(x)$ to be the most frequent label (where g is the smoothed classifier)
- g comes with robustness guarantees!

Creating random perturbations

Given an input x , generate input $x + \epsilon$ where ϵ is noise sampled a Gaussian distribution with mean 0 and variance σ^2 , i.e., $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$



Examples of noisy images for CIFAR-10

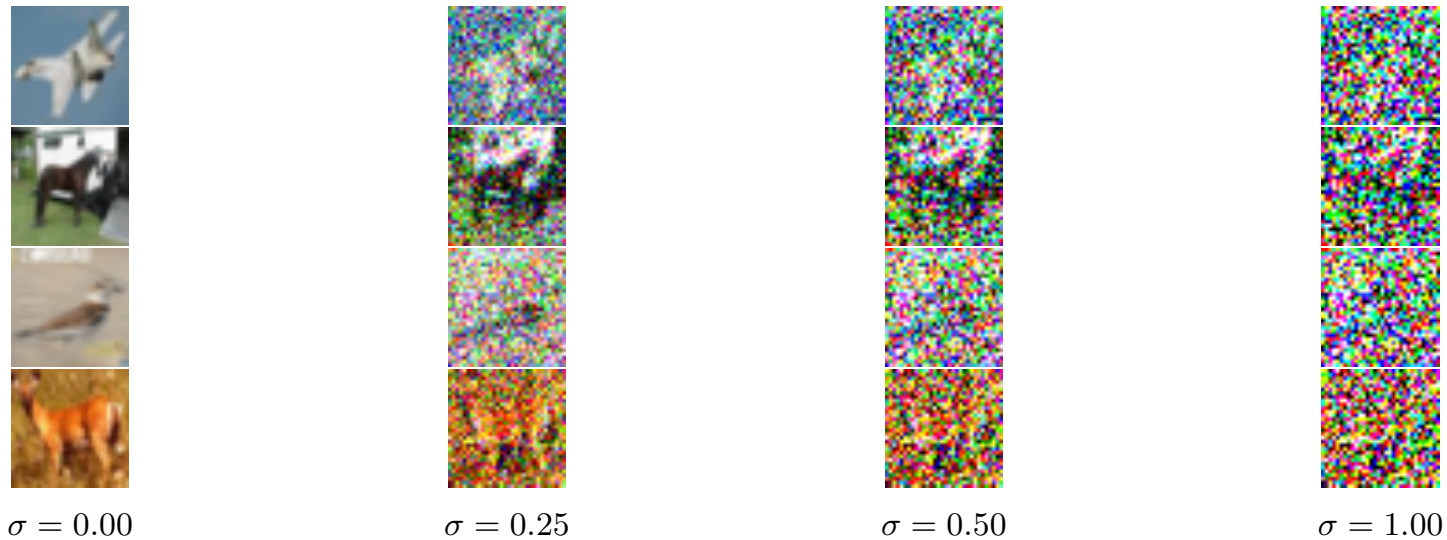


Figure 17. CIFAR-10 images additively corrupted by varying levels of Gaussian noise $\mathcal{N}(0, \sigma^2 I)$. Pixel values greater than 1.0 (=255) or less than 0.0 (=0) were clipped to 1.0 or 0.0.

Example of noisy images for ImageNet

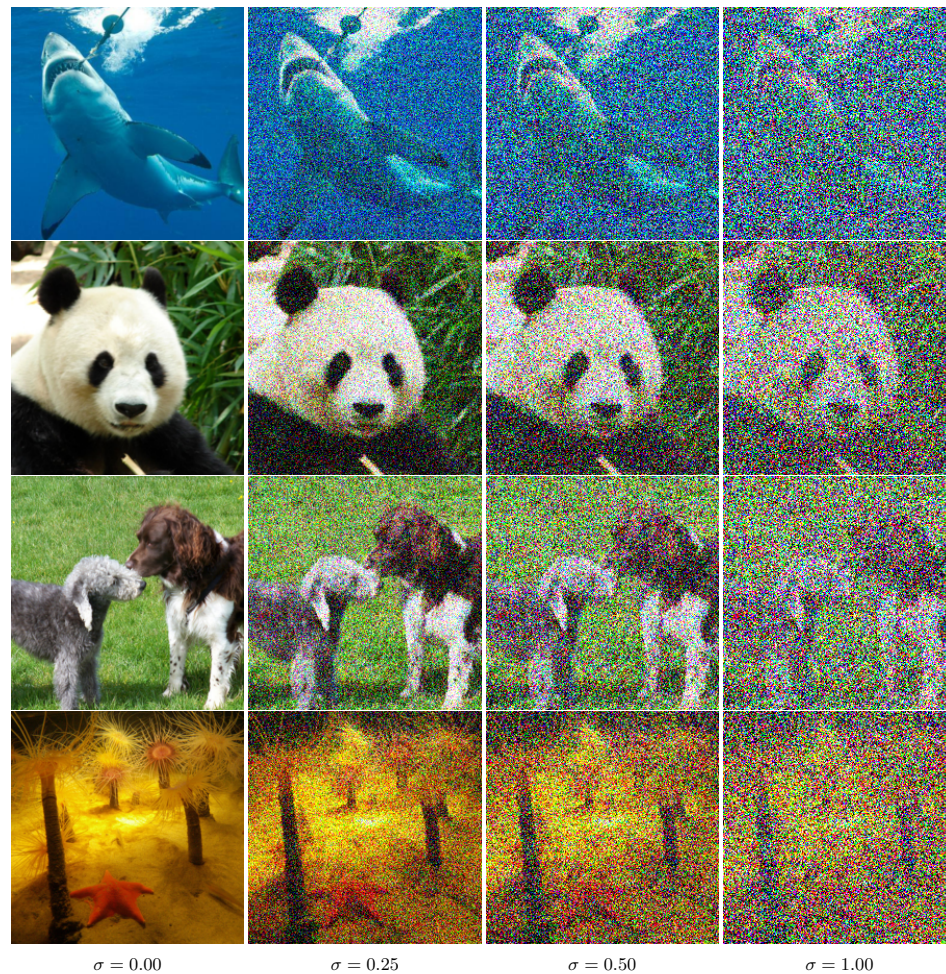


Figure 18. ImageNet images additively corrupted by varying levels of Gaussian noise $\mathcal{N}(0, \sigma^2 I)$. Pixel values greater than 1.0 (≈ 255) or less than 0.0 (≈ 0) were clipped to 1.0 or 0.0.

Randomized smoothing, formally

$$g(x) = \arg \max_{c \in \mathcal{Y}} \mathbb{P}(f(x + \varepsilon) = c)$$

where $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$

where f is the base classifier,

$\mathcal{Y}$ is the set of labels,

g is the smoother classifier

g is not computable!!

Implementing g via Monte Carlo Sampling

To predict the label at input x using smoothed classifier g :

- (1) Create n versions of x by adding noise $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$
- (2) Compute labels on all perturbed inputs $x + \epsilon$ using base classifier f
- (3) Identify top two classes $\hat{c}_A$ and $\hat{c}_B$ with highest number (n_A and n_B , respectively) of predictions
- (4) If n_A is much greater than n_B , return $\hat{c}_A$ as prediction else abstain

evaluate g at x

function PREDICT(f, σ, x, n, α)

 counts $\leftarrow$ SAMPLEUNDERNOISE(f, x, n, σ)

$\hat{c}_A, \hat{c}_B \leftarrow$ top two indices in counts

$n_A, n_B \leftarrow$ counts[$\hat{c}_A$], counts[$\hat{c}_B$]

if BINOMPVALUE($n_A, n_A + n_B, 0.5$) $\leq \alpha$ **return** $\hat{c}_A$

else return ABSTAIN

Proposition 1. *With probability at least $1 - \alpha$ over the randomness in PREDICT, PREDICT will either abstain or return $g(x)$. (Equivalently: the probability that PREDICT returns a class other than $g(x)$ is at most α .)*

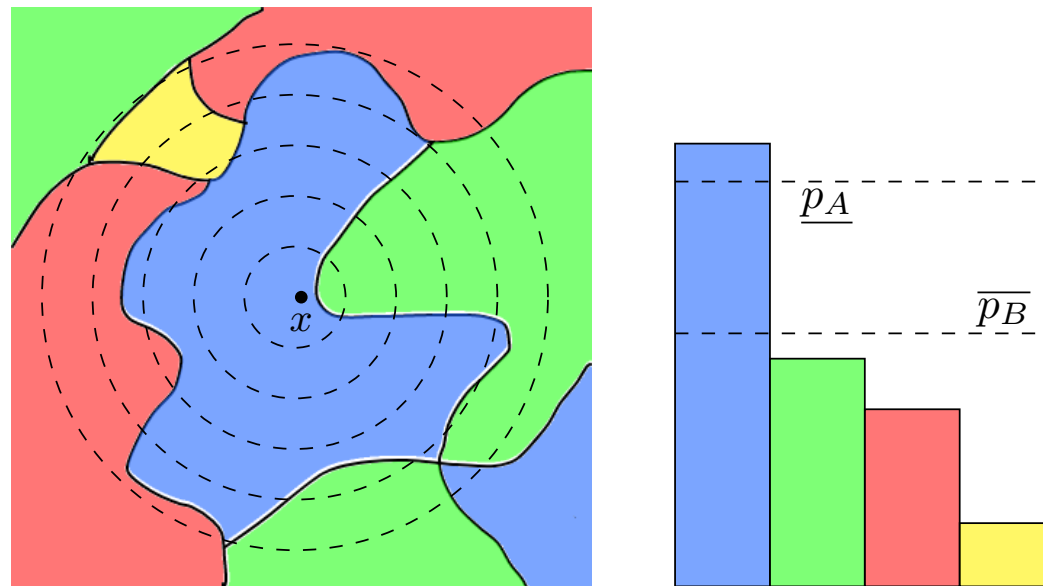
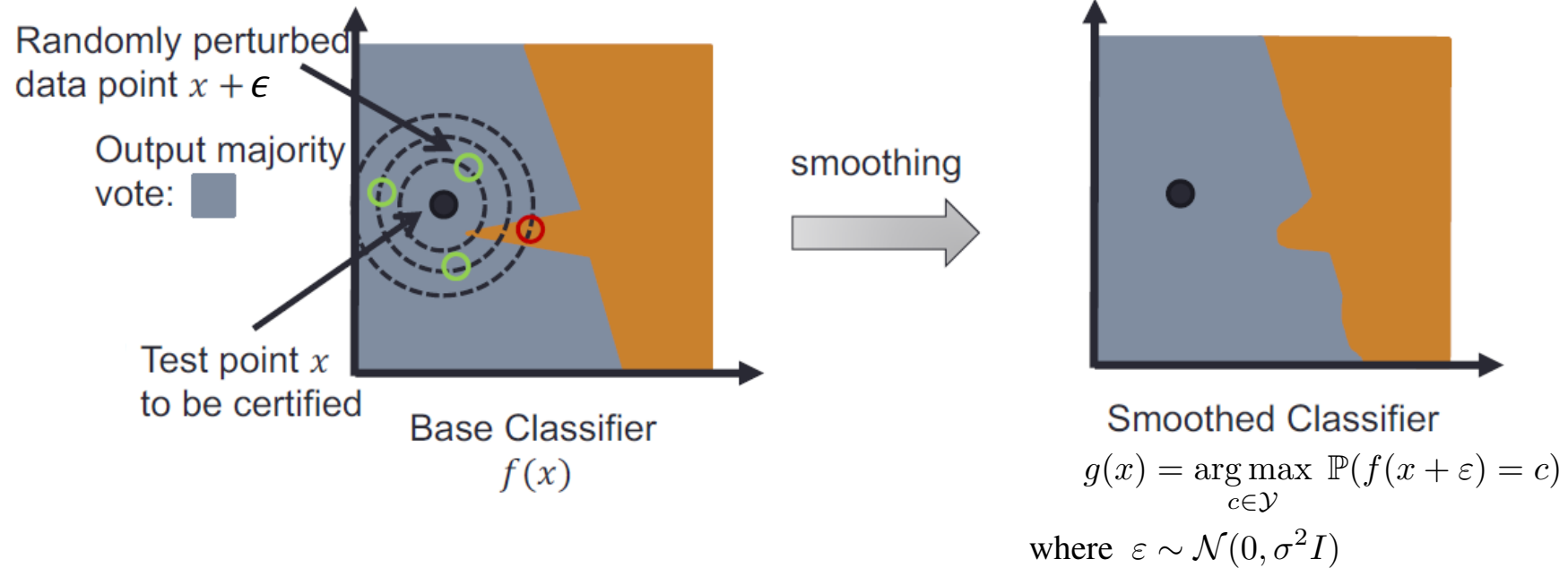


Figure 1. Evaluating the smoothed classifier at an input x . **Left:** the decision regions of the base classifier f are drawn in different colors. The dotted lines are the level sets of the distribution $\mathcal{N}(x, \sigma^2 I)$. **Right:** the distribution $f(\mathcal{N}(x, \sigma^2 I))$. As discussed below, $\underline{p}_A$ is a lower bound on the probability of the top class and $\overline{p}_B$ is an upper bound on the probability of each other class. Here, $g(x)$ is “blue.”

Illustrating effect of smoothing



Robustness guarantee of Randomized Smoothing

$$g(x) = \arg \max_{c \in \mathcal{Y}} \mathbb{P}(f(x + \varepsilon) = c) \quad (1)$$

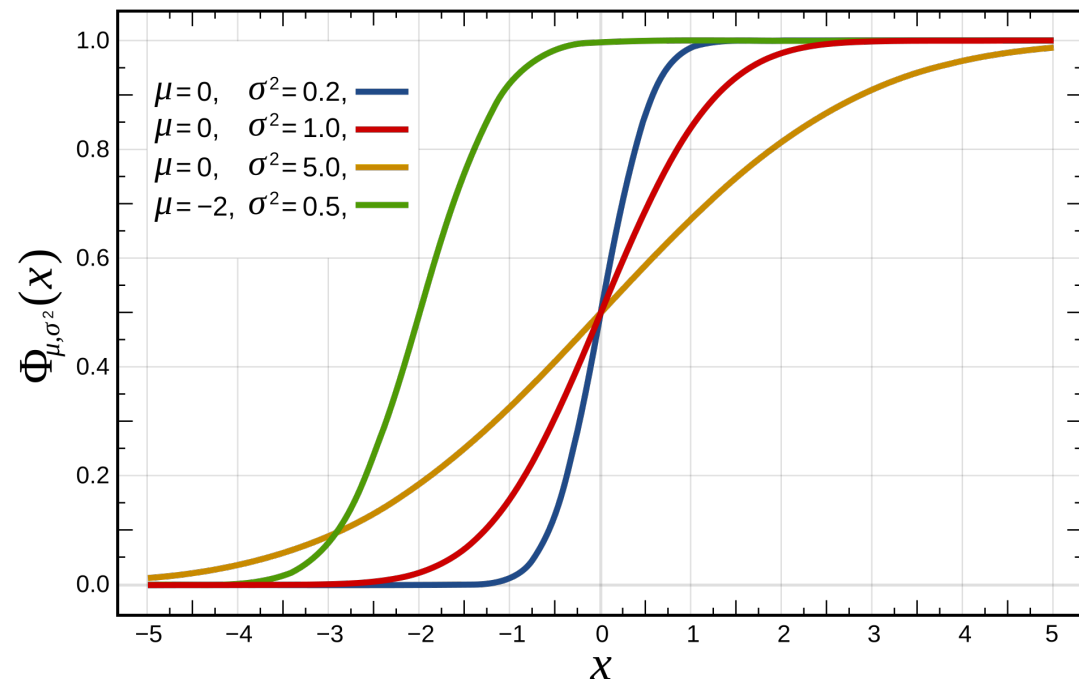
where $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$

Theorem 1. Let $f : \mathbb{R}^d \rightarrow \mathcal{Y}$ be any deterministic or random function, and let $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$. Let g be defined as in (1). Suppose $c_A \in \mathcal{Y}$ and $\underline{p}_A, \overline{p}_B \in [0, 1]$ satisfy:

$$\mathbb{P}(f(x + \varepsilon) = c_A) \geq \underline{p}_A \geq \overline{p}_B \geq \max_{c \neq c_A} \mathbb{P}(f(x + \varepsilon) = c) \quad (2)$$

Then $g(x + \delta) = c_A$ for all $\|\delta\|_2 < R$, where

$$R = \frac{\sigma}{2} (\Phi^{-1}(\underline{p}_A) - \Phi^{-1}(\overline{p}_B)) \quad (3)$$



Gaussian CDFs

Implementing g via Monte Carlo Sampling

certify the robustness of g around x

function CERTIFY($f, \sigma, x, n_0, n, \alpha$)

counts0 $\leftarrow$ SAMPLEUNDERNOISE(f, x, n_0, σ)

$\hat{c}_A \leftarrow$ top index in counts0

counts $\leftarrow$ SAMPLEUNDERNOISE(f, x, n, σ)

$\underline{p}_A \leftarrow$ LOWERCONFBOUND(counts[$\hat{c}_A$], $n, 1 - \alpha$)

if $\underline{p}_A > \frac{1}{2}$ **return** prediction $\hat{c}_A$ and radius $\sigma \Phi^{-1}(\underline{p}_A)$

else return ABSTAIN

- `SampleUnderNoise( $f, x, n, \sigma$ )` samples n values of noise ϵ from the distribution $\mathcal{N}(0, \sigma^2 I)$, evaluates $f(x + \epsilon)$, and returns a vector of class counts
- `LowerConfBound( $k, n, 1 - \alpha$ )` returns one-sided $(1 - \alpha)$ lower interval for the Binomial parameter p given the sample $k \sim \text{Binomial}(n, p)$

Implementing g via Monte Carlo Sampling

certify the robustness of g around x

function CERTIFY($f, \sigma, x, n_0, n, \alpha$)

counts0 $\leftarrow$ SAMPLEUNDERNOISE(f, x, n_0, σ)

$\hat{c}_A \leftarrow$ top index in counts0

counts $\leftarrow$ SAMPLEUNDERNOISE(f, x, n, σ)

$\underline{p}_A \leftarrow$ LOWERCONFBOUND(counts[$\hat{c}_A$], $n, 1 - \alpha$)

if $\underline{p}_A > \frac{1}{2}$ **return** prediction $\hat{c}_A$ and radius $\sigma \Phi^{-1}(\underline{p}_A)$

else return ABSTAIN

Proposition 2. *With probability at least $1 - \alpha$ over the randomness in CERTIFY, if CERTIFY returns a class $\hat{c}_A$ and a radius R (i.e. does not abstain), then g predicts $\hat{c}_A$ within radius R around x : $g(x + \delta) = \hat{c}_A \ \forall \ \|\delta\|_2 < R$.*

Comments on Randomized Smoothing

- Gaussian noise for ℓ_2 perturbations (see [Randomized Smoothing of All Shapes and Sizes](#) for other perturbations)
- Need to train f with noisy images for technique to work well in practice
- Method works for arbitrary f including complex neural networks
- Robustness radius R depends on noise parameter σ and separation between probabilities of top two classes
- Accuracy-robustness tradeoff

Noise vs Resolution

Can use stronger noise for higher resolution images

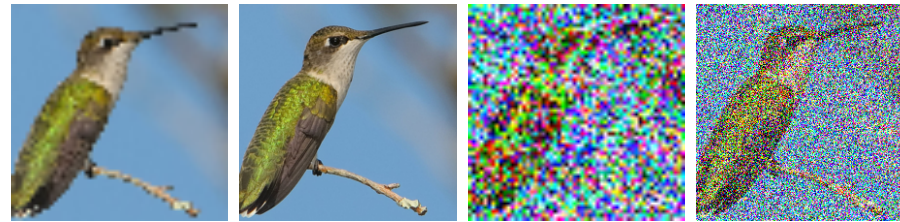
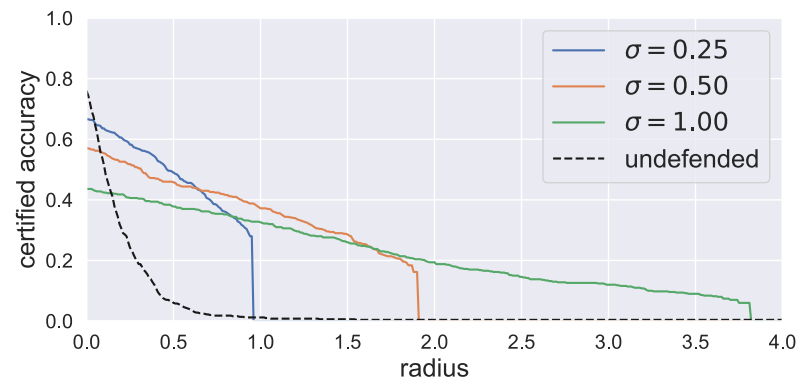
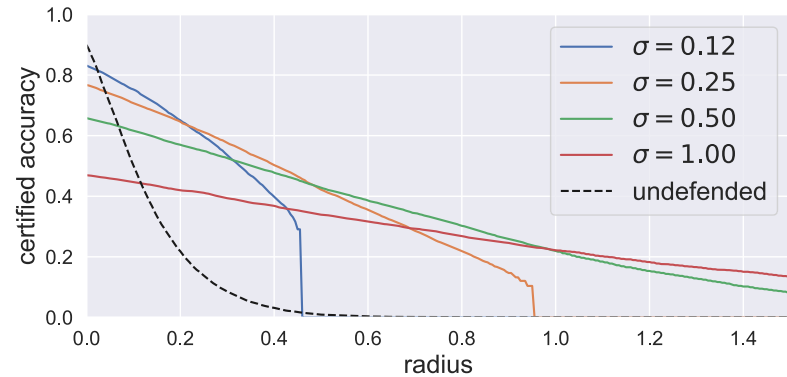


Figure 4. Left to right: clean 56 x 56 image, clean 224 x 224 image, noisy 56 x 56 image ($\sigma = 0.5$), noisy 224 x 224 image ($\sigma = 0.5$).

Empirical evaluation



Notice:

- Effect of radius on certified accuracy
- Effect of σ on certified accuracy
- Randomized smoothing can be very expensive!

$$\text{Certified Accuracy} = \frac{\text{\# of test samples where model is accurate and certified locally robust}}{\text{\# of test samples}}$$